

Mohamad Latifian

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Research Interests

My research lies at the intersection of theoretical computer science, artificial intelligence, and economics, with an emphasis on algorithms and computational complexity. In particular, I am quite interested in problems related to social choice theory, fair division, algorithmic game theory and mechanism design.

Education and Academic Positions

Postdoctoral Research Associate

University of Edinburgh, UK

Host: Aris Filos-Ratsikas

Jul. 2024 – present

Ph.D. in Computer Science

University of Toronto, Canada

Advisors: Nisarg Shah and Allan Borodin

Thesis: Decoding Distortion: Pushing the Limits of Optimal Collective Decision-Making

GPA: 4+/4

Sep. 2020 – Sep. 2024

M.Sc. in Algorithms and Computation

Sharif University of Technology, Iran

Advisor: Mohammad Ghodsi

Thesis: Considering the Effect of Abstention in the Distortion Value.

GPA: 19.59/20

Sep. 2017 – Sep. 2019

B.Sc. in Computer Engineering

Sharif University of Technology, Iran

Advisor: Hamid Zarrabi-zadeh

Thesis: Hamiltonian Path in Solid Grid Graphs.

GPA: 17.22/20

Sep. 2012 – Jul. 2017

Conference Publications

(Authors in alphabetical order)

- C17. *Metric Distortion with Preference Intensities*. M. Abbaszadeh, A. Ansarifard, M. Latifian, and M. Seddighin. Proceedings of the 40th AAAI Conference on Artificial Intelligence (AAAI) (Forthcoming).
- C16. *Optimal Metric Distortion for Matching on the Line*. A. Filos-Ratsikas, V. Gkatzelis, M. Latifian, E. Rewinski, and A. A. Voudouris. 34rd International Joint Conference on Artificial Intelligence (IJCAI), pp. 1–25, 2025.
- C15. *Towards Fair and Efficient Public Transportation: A Bus Stop Model*. M. Bullinger, E. Elkind, and M. Latifian. Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 427–435, 2025.
- C14. *Temporal Fair Division of Indivisible Items*. E. Elkind, A. Lam, M. Latifian, T. Y. Neoh, and N. Teh. Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 676–685, 2025.

- C13. *Fair Division with Market Values*. S. Barman, S. Ebadian, M. Latifian, and N. Shah. Proceedings of the 39th AAAI Conference on Artificial Intelligence (**AAAI**), volume 39, pp. 13589–13596, 2025.
- C12. *The Distortion of Public-spirited Participatory Budgeting*. M. Bedaywi, B. Flanigan, M. Latifian, and N. Shah. Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI**), volume 39, pp. 13605–13613, 2025.
- C11. *The Distortion of Threshold Approval Matching*. M. Latifian and A. A. Voudouris. Proc. of the 33rd International Joint Conference on Artificial Intelligence (**IJCAI**), pp. 2851–2859, 2024.
- C10. *Computational Aspects of Distortion*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah. Proc. of the 23rd International Conference on Autonomous Agents and Multi-Agent Systems (**AAMAS**), pp. 1–14, 2023.
- C9. *Explainable and Efficient Randomized Voting Rules*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah. Proc. of 37th Advances in Neural Information Processing Systems (**NeurIPS**), 36, 2024.
- C8. *Best of Both Distortion Worlds*. V. Gkatzelis, M. Latifian, and N. Shah. Proc. of the 24th ACM Conference on Economics and Computation (**EC**), pp. 738–758, 2023.
- C7. *The Distortion of Approval Voting with Runoff*. S. Ebadian, M. Latifian, and N. Shah. Proc. of the 22nd International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), pp. 1752–1760, 2023.
- C6. *Voting with Preference Intensities*. A. Kahng, M. Latifian, and N. Shah. Proc. of 37th AAAI Conference on Artificial Intelligence (**AAAI**), pp. 5697–5704, 2023.
- C5. *Distortion in Voting with Top- t Preferences*. A. Borodin, D. Halpern, M. Latifian, and N. Shah. Proc. of the 31st International Joint Conference on Artificial Intelligence (**IJCAI**), pp. 116–122, 2022.
- C4. *Almost Envy-freeness, Envy-rank, and Nash Social Welfare Matchings*. A. Farhadi, M. Hajiaghayi, M. Latifian, M. Seddighin, and H. Yami. Proc. of the 35th AAAI Conference on Artificial Intelligence (**AAAI**), pp. 5355–5362, 2021.
- C3. *Resource-aware Protocols for Network Cost-sharing Games*. G. Christodoulou, V. Gkatzelis, M. Latifian, and A. Sgouritsa. Proc. of the 21st ACM conference on economics and computation (**EC**), pp. 81–107, 2020.
- C2. *On the Distortion Value of the Elections with Abstention*. M. Ghodsi, M. Latifian, and M. Seddighin. Proc. of the 33rd AAAI Conference on Artificial Intelligence (**AAAI**), pp. 1981–1988, 2019.
- C1. *Rent Division Among Groups*. M. Ghodsi, M. Latifian, A. Mohammadi, S. Moradian, and M. Seddighin. Proc. of the 12th International Conference on Combinatorial Optimization and Applications (**COCOA**), pp. 577–591, 2018.

Journal Publications

- J2. *The Distortion of Threshold Approval Matching*. M. Latifian and A. A. Voudouris. Autonomous Agents and Multi-Agent Systems, 39(2):pp. 1–21, 2025.
- J1. *On the Distortion Value of Elections with Abstention*. M. Seddighin, M. Latifian, and M. Ghodsi. Journal of Artificial Intelligence Research, 70:pp. 567–595, 2021.

Working Papers

(Authors in alphabetical order)

- M5. *Envy-free Cake Cutting for Five Agents*. A. Filos-Ratsikas, A. Hollender, and M. Latifian.
- M4. *Metric Distortion with Graph Objectives*. A. Filos-Ratsikas, M. H. Ghasemi, M. Latifian, and M. Sharify.
- M3. *Distortion of Thruthful Matching Mechanisms*. V. Gkatzelis, M. Latifian, and E. Rewinski.
- M2. *A Deeper Look at Distortion in Impartial Culture Electorates*. H. Alipour, K. Joolaei, M. Latifian, S. Mizanian, and M. Seddighin.
- M1. *Social Deliberation: Democratic Deliberation Through Unbiased Social Interactions*. D. Afazeli, M. Latifian, and M. Seddighin. (Under Submission)

Academic Visits

Long Term

University of Oxford, UK. Sep. – Nov. 2023
 Host: Edith Elkind

Max Planck Institute for Informatics, Germany. May – Jun. 2019
 Host: Alkmini Sgouritsa
 Work title: *Resource-Aware Protocols for Network Cost-Sharing Games* ([C3.](#))

Short Term

University of Oxford, UK. Sep. 2025
 Host: Alexandros Hollender

University of Edinburgh, UK. Nov. 2023
 Host: Aris Filos-Ratsikas

University of Essex, UK. Oct. 2023
 Host: Alexandros Voudouris

Professional Service

- **PC Member:** EC '26, AAAI '26, EC '25, AAMAS '25, ICLR '25, AAAI '25, NeurIPS '24, IJCAI '24, AAMAS '24, AAAI '24, AAAI '21
- **Journal Reviewer:** Autonomous Agents and Multi-Agent System (JAAMAS), Artificial Intelligence (AIJ), Journal of Artificial Intelligence Research (JAIR)
- **Conference Reviewer:** STOC '25

Invited Talks

- T7. *Fair Division with Market Values*, Workshop on Fair Resource Allocation: Concepts, Algorithms and Complexity (FaiRACAC '26), held in conjunction with ICALP 2025.
- T6. *Distortion in Social Choice Problems*, University of Edinburgh, EconCS Group Meetings (Oct. 2024).
- T5. *Fair Division with Market Values*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFSCS) Lab Lunch (Oct. 2024).

- T4. *How to Make Efficient (Collective) Decisions*, Tehran Institute for Advanced Studies (Sep. 2024).
- T3. *Best of Both Distortion Worlds*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFSCS) Seminar (Nov. 2023).
- T2. *Best of Both Distortion Worlds*, University of Essex, The Centre for Computational Finance and Economic Agents (CCFEA) Seminar Series (Oct. 2023)
- T1. *Resource-aware Protocols for Network Cost-sharing Games*, The Drexel EconCS Research Group Meetings (Jun. 2020).

Teaching Experience

Guest Lecturer

Oct. 2025

University of Edinburgh

Algorithms and Data Structures ([Course web page](#))

Teaching Assistant

University of Toronto

- Algorithm Design, Analysis & Complexity Fall '23
- Data Structures and Analysis Winter '23
- Algorithmic Game Theory and Mechanism Design Fall '22

Teaching Assistant

(* := Lead TA)

Sharif University of Technology

- Algorithmic Game Theory Fall '17, Spring '19*
- Design and Analysis of Algorithms Fall '14, Fall '18*
- Theory of Languages and Automata Fall '14, Spring '15, Fall '15, Spring '16
- Data Structures and Algorithms Spring '15

Member of the Scientific Committee

Fall 2016

4th Sharif National Students Competition,
Sharif University of Technology.

Head of the Scientific Committee

Aug. 2013 and Aug. 2014

13th & 14th Mehregan Mathematic Workshop,
Shahid Ejeii High School.

Teaching Special Topics in Mathematics

2011-2013

Teaching high-school students preparing for Olympiad in Informatics.

Topics included Algorithms, Graph Theory, Problem-Solving Strategies, and Programming.

Supervision Experience

Mohammad Hossein Ghasemi and Mohammad Sharify (Co-supervised with A. Filos-Ratsikas)

Undergraduate students, Sharif University of Technology

Aug. 2025 – present

Title: *Metric Distortion with Graph Objectives*

Hamidreza Alipour, Kiarash Joolaei, Sepehr Mizanian

Undergraduate students, Sharif University of Technology

Nov. 2024 – present

Title: *A Deeper Look at Distortion in Impartial Culture Electorates*

Dana Afazeli

Undergraduate student, Sharif University of Technology

Title: *Social Deliberation: Democratic Deliberation Through Unbiased Social Interactions*

Aug. 2024 – present

Mehrad Abbaszadeh and Ali Ansarifar (Co-supervised with M. Seddighin)

Undergraduate students, Sharif University of Technology

Title: *Metric Distortion with Preference Intensities (To appear in AAAI '26)*

Oct. 2023 – Aug. 2025

Honors and Awards

- **C.C. Gotlieb (Kelly) Graduate Fellowship** in the Department of Computer Science, University of Toronto 2023
- **Silver medal** in the 21st Iranian National Olympiad in Informatics (INOI). 2011
- **Ranked 2nd and 7th** in Nationwide Graduate Entrance Exam, in Algorithms and Computation and Artificial Intelligence majors respectively. 2017
- **Ranked 1st** in cumulative GPA among 97 M.Sc. students of the department, Sharif University of Technology. 2019
- **Ranked 147th** out of 300,000 competitors in the national university entrance exam. 2012
- Scholarship from Iranian National Elites Foundation 2012 – 2019

Activities

- **Elected Member & President** of the Students Scientific Chapter (SSC) Computer Engineering Department of Sharif University of Technology. 2014 – 2015
- **Chief of Staff** of 17th & 18th International Collegiate Programming Contest (ICPC) Asia Region - Tehran Site. 2015 – 2016
- **Chief of Staff** of 6th & 7th AI Challenge 2014 – 2015
An event where AI agents written in Java, C++, or Python compete to pass the game field and overcome other agents.